

RHEINE BENTLAGE, Germany

WMO: 103060

Lat: 52.2887N

Lon: 7.3867E

Elev: 39

StdP: 100.85

Time Zone: 1.00 (EUC)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-5.8	-10.8	1.5	-7.2	-8.6	1.8	-4.7	11.2	5.8	9.9	6.6	2.8	60	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.0	19.8	27.8	19.0	25.7	18.0	20.8	27.7	19.8	25.8	18.7	24.2	3.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	23.0	17.6	12.7	22.1	16.7	12.0	21.2	60.2	28.2	56.5	26.0	53.2	24.4	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.5	DB	-10.8	34.0	3.8	1.4	-13.5	35.0	-15.8	35.9	-17.9	36.7	-20.7	37.7	(4)
(5)			WB	-11.2	22.7	3.8	1.3	-13.9	23.7	-16.2	24.5	-18.3	25.2	-21.1	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	2.5	3.4	6.1	9.5	13.4	16.1	18.3	17.9	14.7	10.7	6.5	3.6	(6)
(7)		DBStd	6.68	4.55	4.16	3.46	3.72	3.60	3.33	3.24	3.05	2.87	3.45	3.57	4.51	(7)
(8)		HDD10.0	964	234	187	128	54	9	0	0	0	1	34	114	202	(8)
(9)		HDD18.3	3052	492	419	380	265	158	84	42	45	114	238	356	458	(9)
(10)		CDD10.0	1058	1	1	6	39	114	184	258	246	143	54	8	3	(10)
(11)		CDD18.3	102	0	0	0	0	5	17	41	33	6	0	0	0	(11)
(12)		CDH23.3	1037	0	0	0	11	67	182	406	313	58	1	0	0	(12)
(13)		CDH26.7	308	0	0	0	1	12	46	135	104	10	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.5	3.5	3.5	3.1	2.9	2.8	2.8	2.6	2.6	2.8	3.0	3.2	(14)
(15)	Precipitation	PrecAvg	785	70	57	54	45	57	71	79	76	69	65	70	73	(15)
(16)		PrecMax	1060	126	129	126	95	122	160	170	202	151	216	138	170	(16)
(17)		PrecMin	647	7	19	10	2	29	18	23	21	4	15	3	23	(17)
(18)		PrecStd	121	33	31	32	20	25	34	35	43	36	37	32	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.8	19.0	25.1	28.5	31.2	33.1	32.2	28.1	21.8	16.2	13.6	(19)
(20)			MCWB	10.4	9.2	11.5	14.4	18.1	20.8	20.7	20.1	19.2	16.6	13.3	11.4	(20)
(21)		2%	DB	11.1	11.8	15.9	21.3	25.0	27.7	30.2	29.7	25.0	19.0	14.1	12.2	(21)
(22)			MCWB	9.2	8.8	10.3	13.4	16.0	18.8	20.1	19.4	18.2	15.2	12.1	10.4	(22)
(23)		5%	DB	9.8	10.3	13.4	18.9	23.0	25.2	27.7	26.9	22.1	17.1	12.9	11.0	(23)
(24)			MCWB	8.2	8.1	9.3	12.3	15.1	17.6	19.3	18.9	16.6	14.2	11.0	10.1	(24)
(25)		10%	DB	8.3	9.0	11.7	16.4	20.7	23.0	25.2	24.2	20.0	15.8	11.6	9.9	(25)
(26)			MCWB	6.8	7.2	8.4	11.1	14.2	16.5	18.1	17.6	15.5	13.4	9.8	8.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.5	12.6	15.9	18.8	21.6	22.1	21.7	20.1	17.1	13.7	11.7	(27)
(28)			MCDWB	12.2	12.4	16.9	21.9	25.6	28.0	30.0	29.4	26.2	20.9	15.4	12.8	(28)
(29)		2%	WB	9.6	9.3	11.0	14.2	17.4	19.9	21.0	20.4	18.6	15.7	12.4	11.1	(29)
(30)			MCDWB	10.9	11.1	14.6	19.5	23.2	26.1	27.9	27.3	23.7	18.3	14.1	11.8	(30)
(31)		5%	WB	8.1	8.3	9.9	13.1	15.9	18.5	20.0	19.2	17.3	14.7	11.3	10.2	(31)
(32)			MCDWB	9.5	10.1	13.0	17.7	20.9	23.8	25.8	25.2	21.3	16.7	12.8	11.4	(32)
(33)		10%	WB	6.9	7.2	8.7	11.7	14.7	17.2	18.9	18.2	16.1	13.6	10.0	8.5	(33)
(34)			MCDWB	8.3	8.9	11.2	15.8	19.4	21.7	24.1	23.3	19.2	15.6	11.3	9.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.6	7.7	9.8	10.0	10.2	9.9	9.0	7.4	5.3	4.4	(35)	
(36)			MCDBR	5.8	6.9	11.5	13.8	13.8	14.2	14.6	14.5	12.7	9.1	6.4	4.4	(36)
(37)			MCWBR	4.7	5.0	7.1	7.3	6.8	7.0	6.3	6.4	6.8	5.9	5.0	3.8	(37)
(38)		5% WB	MCDWR	5.2	5.8	9.9	11.7	11.9	12.2	12.8	12.2	11.1	8.2	5.9	3.9	(38)
(39)			MCWBR	4.7	4.7	6.6	6.6	6.6	6.6	5.8	6.1	6.3	5.7	4.8	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.346	0.376	0.410	0.412	0.410	0.422	0.415	0.389	0.361	0.341	0.320	(40)	
(41)		taud	2.358	2.341	2.270	2.219	2.247	2.271	2.248	2.287	2.350	2.408	2.374	2.343	(41)	
(42)		Ebn at Noon	664	760	806	823	840	844	825	809	786	734	637	599	(42)	
(43)		Edh at Noon	61	82	107	126	129	127	128	117	98	76	60	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.68	1.35	2.48	4.02	4.90	5.20	4.99	4.19	3.05	1.78	0.82	0.53	(44)	
(45)		RadStd	0.08	0.24	0.34	0.43	0.44	0.40	0.63	0.35	0.35	0.17	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page